

Example 10.4

$$\text{ARIMA}(0, 0, 1) \times (1, 0, 1)_{12}$$

$$\text{Period} = 12$$

$$\text{Seasonal components: } p = 1, d = 0, q = 1$$

$$\text{Non-seasonal components: } p = 0, d = 0, q = 1$$

$$\phi_p(B) = 1; \quad \Phi_p(B^{12}) = 1 - \Phi_1 B^{12}$$

$$\theta_q(B) = 1 + \theta_1 B; \quad \Theta_q(B^{12}) = 1 + \Theta_1 B^{12}$$

$$\phi_p(B) \Phi_p(B^{12}) X_t = \theta_q(B) \Theta_q(B^{12}) Z_t$$

$$(1 - \Phi_1 B^{12}) X_t = (1 + \theta_1 B)(1 + \Theta_1 B^{12}) Z_t$$

$$\Rightarrow X_t - \Phi_1 X_{t-12} = Z_t + \theta_1 Z_{t-1} + \Theta_1 Z_{t-12} + \theta_1 \Theta_1 Z_{t-13}$$